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|-------------|---------------|------------------|
| Surname     | Centre Number | Candidate Number |
| Other Names |               | 0                |

**GCSE – NEW**

3410UA0-1



S18-3410UA0-1

**CHEMISTRY – Unit 1:**  
**Chemical Substances, Reactions and**  
**Essential Resources**

**HIGHER TIER**

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 45 minutes

**ADDITIONAL MATERIALS**

In addition to this examination paper  
 you will need a calculator and a ruler.

**INSTRUCTIONS TO CANDIDATES**

Use black ink or black ball-point pen. Do not use gel pen  
 or correction fluid.

Write your name, centre number and candidate number  
 in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.  
 If you run out of space, use the additional page at the back  
 of the booklet, taking care to number the question(s) correctly.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

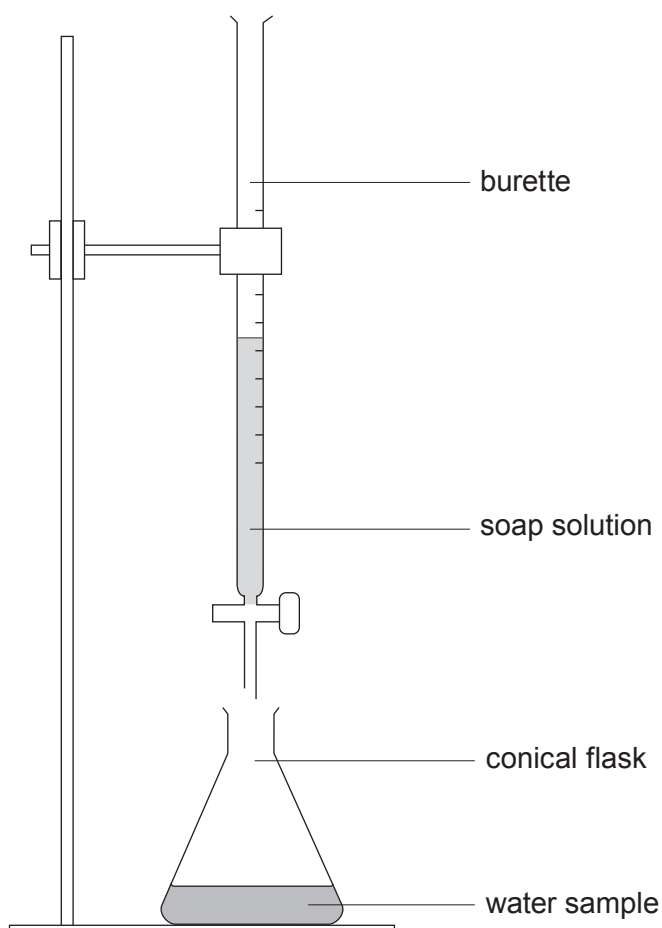
| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 6            |              |
| 2.                      | 9            |              |
| 3.                      | 5            |              |
| 4.                      | 9            |              |
| 5.                      | 11           |              |
| 6.                      | 6            |              |
| 7.                      | 10           |              |
| 8.                      | 10           |              |
| 9.                      | 8            |              |
| 10.                     | 6            |              |
| <b>Total</b>            | <b>80</b>    |              |

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Answer **all** questions.

1. Water samples **A**, **B**, **C** and **D** were tested for hardness using the apparatus shown.



Soap solution was added  $1\text{ cm}^3$  at a time to each sample and the volume required to produce a permanent lather on shaking was recorded. Each sample was tested before and after boiling. The results are shown in the table.

| Water sample | Volume of soap solution required ( $\text{cm}^3$ ) |               |
|--------------|----------------------------------------------------|---------------|
|              | Before boiling                                     | After boiling |
| <b>A</b>     | 1                                                  | 1             |
| <b>B</b>     | 10                                                 | 10            |
| <b>C</b>     | 15                                                 | 1             |
| <b>D</b>     | 15                                                 | 8             |



- (a) (i) State which water sample contains **only** temporary hardness. Explain your answer. [2]

Water sample .....

Explanation .....

- (ii) Give **one** similarity in the composition of temporary and permanent hard water. [1]

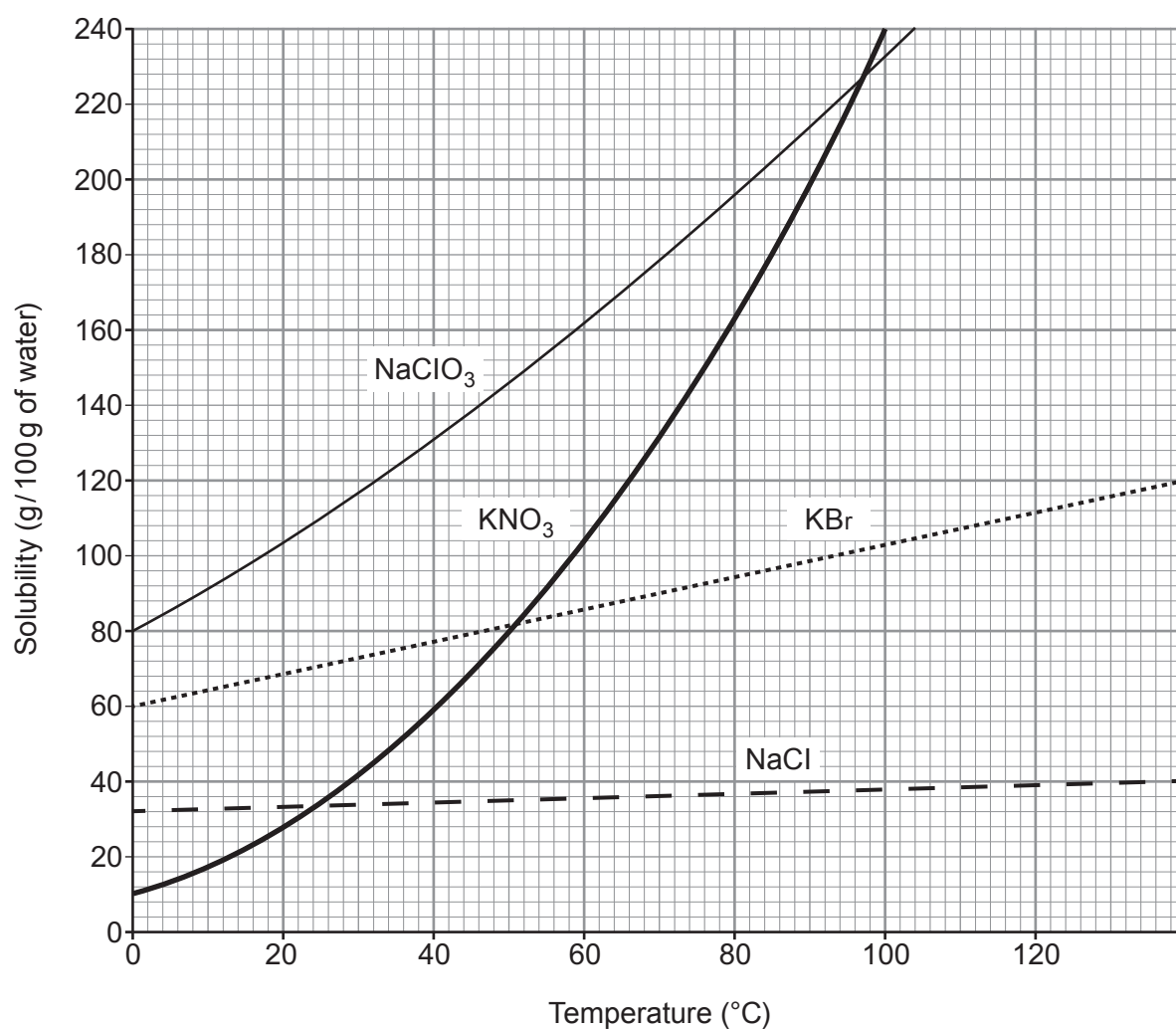
- (b) Discuss the benefits and drawbacks of living in a hard water area. [3]

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6



2. The grid below shows the solubility curves for four ionic compounds.



|                    |                   |
|--------------------|-------------------|
| NaClO <sub>3</sub> | sodium chlorate   |
| KNO <sub>3</sub>   | potassium nitrate |
| KBr                | potassium bromide |
| NaCl               | sodium chloride   |



- (a) (i) Give the temperature at which the solubility of potassium nitrate and potassium bromide is the same. [1]

..... °C

- (ii) Calculate the mass of solid potassium nitrate that would form if a saturated solution in 200 g of water were cooled from 100 °C to 20 °C. [3]

Mass = ..... g

- (iii) Suggest why a student may be surprised at the temperature range shown on the solubility curves. [1]

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- (b) (i) Give the symbols of the **ions** of Group 1 elements present in the compounds shown on the grid. [1]

.....

- (ii) Explain how these ions are formed from their atoms. [2]

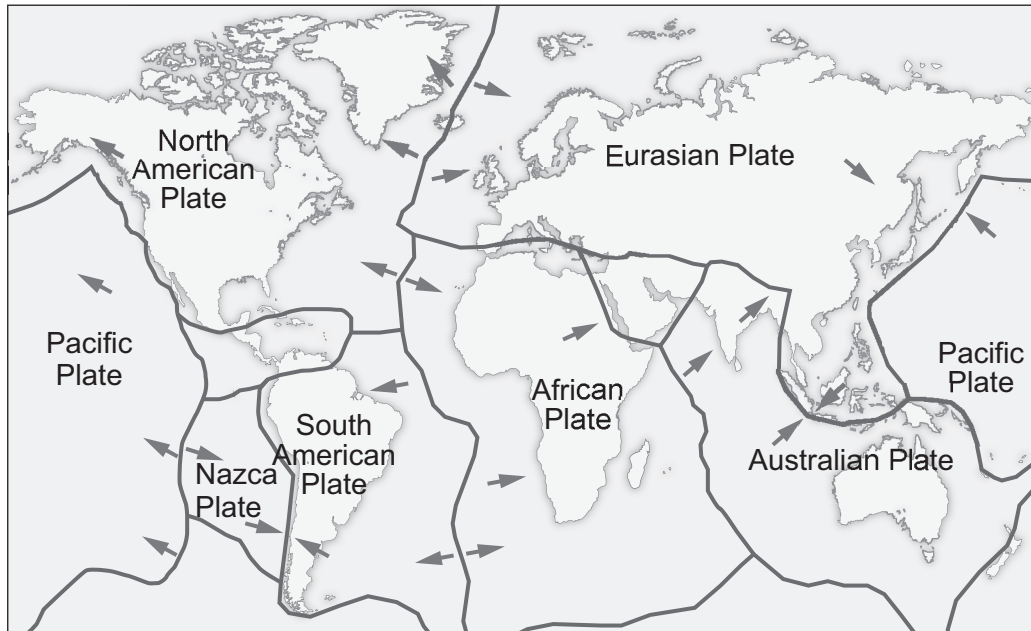
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- (c) Potassium nitrate reacts with aluminium hydroxide to produce aluminium nitrate and potassium hydroxide.

Balance the symbol equation for the reaction taking place. [1]



3. The following diagram shows some of the Earth's tectonic plates and the direction in which they move.



- (a) The boundary between the Nazca and South American plates is a destructive plate boundary. Describe what happens at a destructive boundary. [2]

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- (b) Draw a cross (✕) on the diagram to show a constructive plate boundary. Describe what happens at this boundary. [2]

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- (c) State **one** effect of plates sliding past each other. [1]

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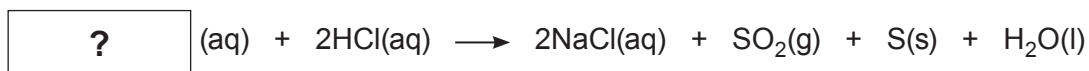


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4. (a) Dilute hydrochloric acid reacts with sodium thiosulfate to make the products shown in the equation.



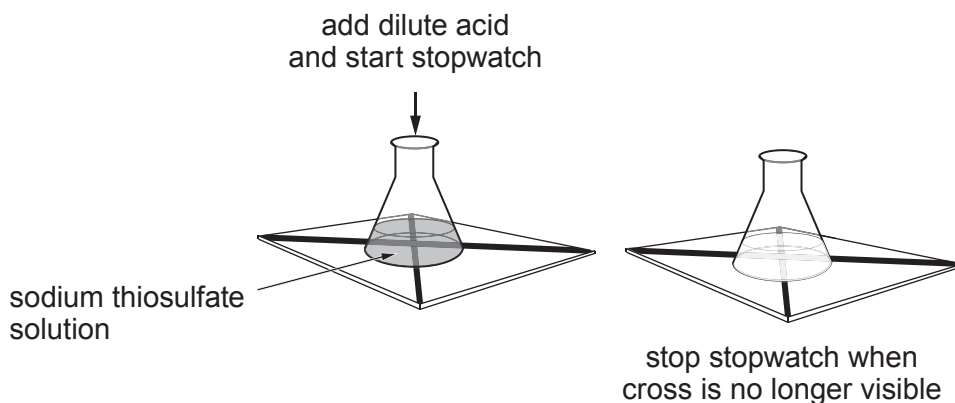
- (i) Use the equation to work out the formula of sodium thiosulfate. [1]

Formula .....

- (ii) The symbol (aq) in the equation tells us that the substances are aqueous. What is meant by this? [1]

.....

- (iii) The rate of this reaction can be studied as shown in the diagram.



- Use information **from the equation** to explain why the cross disappears. [2]

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- (b) A student studied the effect of temperature on the rate of this reaction. He obtained the following results.

| Temperature<br>(°C) | Time taken for cross to disappear (s) |     |     |      |
|---------------------|---------------------------------------|-----|-----|------|
|                     | 1                                     | 2   | 3   | Mean |
| 15                  | 130                                   | 128 | 129 | 129  |
| 30                  | 53                                    | 53  | 53  | 53   |
| 45                  | 21                                    | 29  | 23  | 24.3 |
| 60                  | 7                                     | 7   | 6   | 6.7  |

- (i) Another student said that one of the mean values was incorrect. Identify the incorrect mean. Give your reasoning. [2]

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- (ii) State what conclusion can be drawn about the effect of temperature on the rate of this reaction. Explain your conclusion using particle theory. [3]

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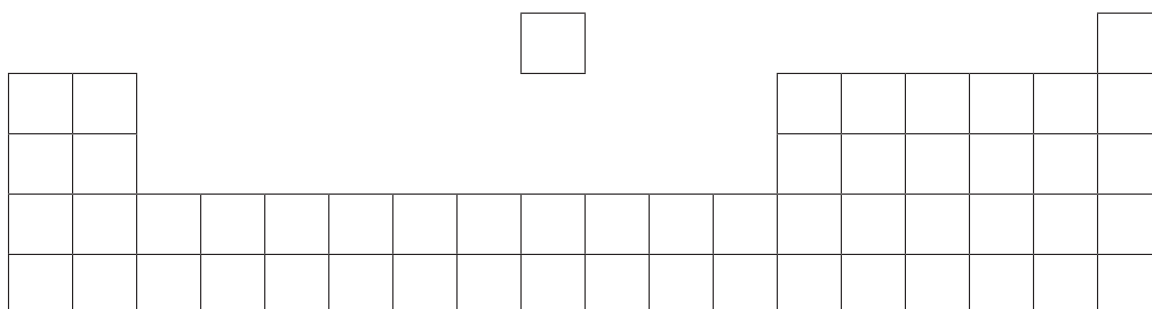
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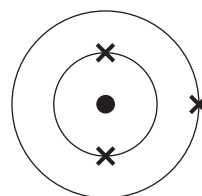


5. The following diagram is an outline of the Periodic Table.



(a) Write letters **A**, **B** and **C** on the **diagram** in the positions of the elements that fit the following descriptions. [3]

**A** the element with the electronic structure



**B** the element in Group 2 and Period 4

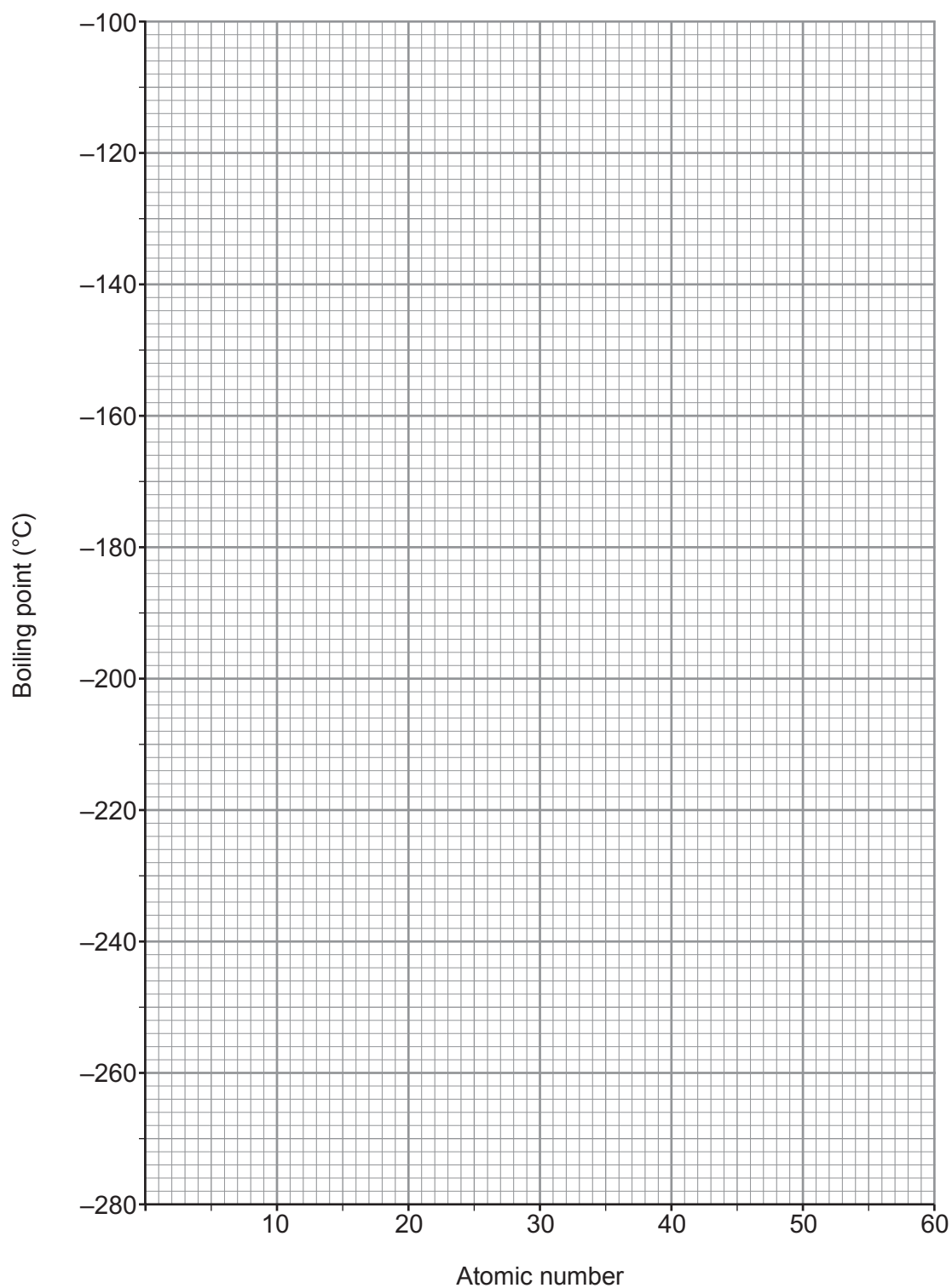
**C** an element that shows both metallic and non-metallic properties

(b) The following table shows the atomic numbers and boiling points of the inert gases.

| Inert gas          | helium | neon | argon | krypton | xenon |
|--------------------|--------|------|-------|---------|-------|
| Atomic number      | 2      | 10   | 18    | 36      | 54    |
| Boiling point (°C) | -269   | -246 | -186  | -153    | -108  |

(i) Plot this data on the grid opposite. Draw a suitable line. [3]





(ii) Describe the trend in boiling point shown on the graph.

[1]

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.....



(c) The following table shows the boiling points of the inert gases in both °C and K.

|                    |        |      |       |         |       |
|--------------------|--------|------|-------|---------|-------|
| Inert gas          | helium | neon | argon | krypton | xenon |
| Boiling point (°C) | −269   | −246 | −186  | −153    | −108  |
| Boiling point (K)  | 4      | 27   | ..... | 120     | 165   |

Use the information in the table to calculate the boiling point of argon in K. [2]

Boiling point = ..... K

(d) Give **one** use of argon. Explain in terms of electronic structure why it is used for this purpose. [2]

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- Describe and explain the sequence of reactions carried out in the laboratory to convert limestone into slaked lime. [6 QER]

6



7. (a) Group 7 elements are known as the halogens. The following table shows the observations made when the first three members of the group react with hydrogen.

| Halogen  | Observations                              |
|----------|-------------------------------------------|
| fluorine | explodes in cold and dark                 |
| chlorine | explodes in sunlight                      |
| bromine  | small explosion when ignited with a flame |

- (i) Use your knowledge of electronic structure to explain why all the halogens react in a similar way and why they react more slowly on going down the group. [3]

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- (ii) Hydrogen fluoride is highly corrosive and can be used to etch glass which is mainly silicon dioxide.

Balance the symbol equation for the reaction between hydrogen fluoride and silicon dioxide. [1]



- (iii) Calcium fluoride reacts with sulfuric acid,  $\text{H}_2\text{SO}_4$ , to produce calcium sulfate and hydrogen fluoride. Give the **symbol** equation for the reaction. [3]

.....



(b) Chlorine reacts with aluminium to produce aluminium chloride.

A sample of aluminium chloride of mass 26.70 g was found to contain 5.45 g of aluminium.  
Calculate the simplest formula of this chloride of aluminium.

You **must** show your working.

[3]

$$A_r(\text{Al}) = 27$$

$$A_r(\text{Cl}) = 35.5$$

Formula .....

|    |
|----|
|    |
| 10 |



8. Sodium is extracted from sodium chloride.

(a) The overall reaction taking place is shown in the equation below.



- (i) When carrying out the reaction 120 kg of sodium chloride was found to produce 38.05 kg of sodium.

Calculate the maximum possible mass of sodium that could be produced and use this figure to calculate the percentage yield of this reaction. [4]

$$A_r(\text{Na}) = 23 \quad A_r(\text{Cl}) = 35.5$$

Maximum possible mass = ..... kg

Percentage yield = ..... %

- (ii) Suggest a reason why the yield is less than 100%. [1]

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- (iii) Suggest why this reaction must be carried out under dry conditions. [1]

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(b) A sample of lithium is found to contain two isotopes.

| Isotope   | Percentage present in sample (%) |
|-----------|----------------------------------|
| lithium-6 | 7.59                             |
| lithium-7 | 92.41                            |

- (i) Calculate the relative atomic mass ( $A_r$ ) of lithium. Give your answer to **three** significant figures. [3]

$$A_r = \frac{(\text{isotope 1 mass} \times \text{abundance}) + (\text{isotope 2 mass} \times \text{abundance})}{100}$$

$A_r =$  .....

- (ii) State the difference between the atomic structures of lithium-6 and lithium-7. [1]

.....  
 .....



9. The following table shows the decomposition temperatures of Group 2 metal carbonates and nitrates.

| Metal     | Temperature at which the carbonate decomposes (°C) | Temperature at which the nitrate decomposes (°C) |
|-----------|----------------------------------------------------|--------------------------------------------------|
| magnesium | 117                                                | 89                                               |
| calcium   | 178                                                | 561                                              |
| strontium | 235                                                | 570                                              |
| barium    | 267                                                | 700                                              |

- (a) Describe the trends in the stabilities of the Group 2 carbonates and nitrates. [3]

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- (b) When a carbonate decomposes it produces carbon dioxide gas. Describe an experiment that could be carried out to show that carbon dioxide gas is produced during the reaction. [2]

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- (c) When calcium nitrate decomposes it forms calcium oxide, oxygen and nitrogen dioxide, NO<sub>2</sub>.

Write a **symbol** equation for the reaction. [3]

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- 10.** Fluorine exists naturally as the fluoride ion. It is found in soil, water, foods and several minerals, such as fluorapatite and fluorite.

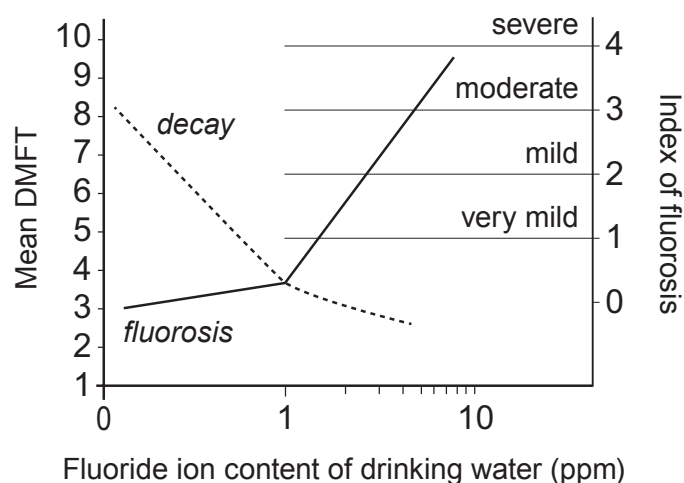
Fluoride ion concentration in seawater averages 1.3 ppm (parts per million). In fresh water, the natural range is typically between 0.01 and 0.3 ppm. In some parts of the world, fresh water contains fluoride ion levels which are dangerous and can lead to health problems.

In the early 1930s, scientists found that people who were brought up in areas with naturally fluoridated water had up to two-thirds fewer cavities compared to those who lived in areas where the water was not fluoridated. Several studies since then have repeatedly shown that when fluoride is added to people's drinking water in areas where natural levels are low, tooth decay decreases.

However, many European countries which do not fluoridate their water do not have a higher incidence of dental decay than countries which do so. It was also found that in Germany and Finland, decay rates either remained stable or continued in their downward trend after they stopped adding fluoride to their drinking water.

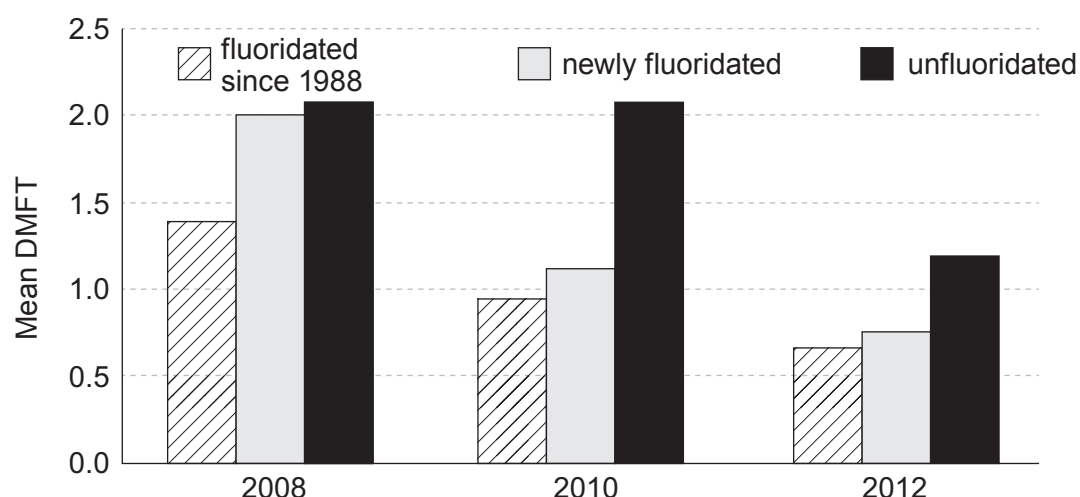
**Figure 1** shows data about the effect of fluoridation of drinking water on the mean number of decayed, missing and filled teeth (DMFT) and the amount of fluorosis seen.

**Figure 2** shows the change in mean DMFT in three regions of Australia over a four year period.



**Figure 1**





**Figure 2**

- (a) Describe the effects of adding varying concentrations of fluoride ions to drinking water. [3]

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- (b) Tick (✓) the statement below that best describes why Germany stopped fluoridating its water supplies. [1]

Ten years after stopping adding fluoride there was no increase in tooth decay

☐

They found that adding fluoride caused fluorosis

☐

Natural water supplies already contain fluoride in a high concentration

☐

Studies showed that areas with no fluoridation did not have higher levels of decay than areas that did fluoridate

☐


- (c) Tick (✓) the box which gives **one** definite conclusion that can be drawn using **only** the data in **Figure 2**. [1]

Fluoridation has no effect on levels of decay

☐

People have reduced their intake of sugary foods over this period

☐

More than one factor affects levels of decay

☐

Fluoridation is the main cause of falling levels of decay

☐

- (d) 'Mass medication' is an argument often given to oppose fluoridation of water supplies. Explain what is meant by the term *mass medication*. [1]

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## FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                  | NEGATIVE IONS |                    |
|---------------|------------------|---------------|--------------------|
| Name          | Formula          | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$ | bromide       | $\text{Br}^-$      |
| ammonium      | $\text{NH}_4^+$  | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$ | chloride      | $\text{Cl}^-$      |
| calcium       | $\text{Ca}^{2+}$ | fluoride      | $\text{F}^-$       |
| copper(II)    | $\text{Cu}^{2+}$ | hydroxide     | $\text{OH}^-$      |
| hydrogen      | $\text{H}^+$     | iodide        | $\text{I}^-$       |
| iron(II)      | $\text{Fe}^{2+}$ | nitrate       | $\text{NO}_3^-$    |
| iron(III)     | $\text{Fe}^{3+}$ | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^+$    | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$ |               |                    |
| nickel        | $\text{Ni}^{2+}$ |               |                    |
| potassium     | $\text{K}^+$     |               |                    |
| silver        | $\text{Ag}^+$    |               |                    |
| sodium        | $\text{Na}^+$    |               |                    |
| zinc          | $\text{Zn}^{2+}$ |               |                    |



2  
1

# Group



4



6

7

0

|                                                                            |                         |                          |                         |                         |                          |                          |                          |                        |                          |                         |                        |                         |                         |                          |                          |                         |                       |  |  |
|----------------------------------------------------------------------------|-------------------------|--------------------------|-------------------------|-------------------------|--------------------------|--------------------------|--------------------------|------------------------|--------------------------|-------------------------|------------------------|-------------------------|-------------------------|--------------------------|--------------------------|-------------------------|-----------------------|--|--|
| <div><div>1H<br/>Hydrogen<br/>1</div><div>4He<br/>Helium<br/>2</div></div> |                         |                          |                         |                         |                          |                          |                          |                        |                          |                         |                        |                         |                         |                          |                          |                         |                       |  |  |
| 7Li<br>Lithium<br>3                                                        | 9Be<br>Beryllium<br>4   |                          |                         |                         |                          |                          |                          |                        |                          | 11B<br>Boron<br>5       | 12C<br>Carbon<br>6     | 14N<br>Nitrogen<br>7    | 16O<br>Oxygen<br>8      | 19F<br>Fluorine<br>9     | 20Ne<br>Neon<br>10       |                         |                       |  |  |
| 23Na<br>Sodium<br>11                                                       | 24Mg<br>Magnesium<br>12 |                          |                         |                         |                          |                          |                          |                        |                          | 27Al<br>Aluminium<br>13 | 28Si<br>Silicon<br>14  | 31P<br>Phosphorus<br>15 | 32S<br>Sulfur<br>16     | 35.5Cl<br>Chlorine<br>17 | 40Ar<br>Argon<br>18      |                         |                       |  |  |
| 39K<br>Potassium<br>19                                                     | 40Ca<br>Calcium<br>20   | 45Sc<br>Scandium<br>21   | 48Ti<br>Titanium<br>22  | 51V<br>Vanadium<br>23   | 52Cr<br>Chromium<br>24   | 55Mn<br>Manganese<br>25  | 56Fe<br>Iron<br>26       | 59Co<br>Cobalt<br>27   | 59Ni<br>Nickel<br>28     | 63.5Cu<br>Copper<br>29  | 65Zn<br>Zinc<br>30     | 70Ga<br>Gallium<br>31   | 73Ge<br>Germanium<br>32 | 75As<br>Arsenic<br>33    | 79Se<br>Selenium<br>34   | 80Br<br>Bromine<br>35   | 84Kr<br>Krypton<br>36 |  |  |
| 86Rb<br>Rubidium<br>37                                                     | 88Sr<br>Strontium<br>38 | 89Y<br>Yttrium<br>39     | 91Zr<br>Zirconium<br>40 | 93Nb<br>Niobium<br>41   | 96Mo<br>Molybdenum<br>42 | 99Tc<br>Technetium<br>43 | 101Ru<br>Ruthenium<br>44 | 103Rh<br>Rhodium<br>45 | 106Pd<br>Palladium<br>46 | 108Ag<br>Silver<br>47   | 112Cd<br>Cadmium<br>48 | 115In<br>Indium<br>49   | 119Sn<br>Tin<br>50      | 122Sb<br>Antimony<br>51  | 128Te<br>Tellurium<br>52 | 127I<br>Iodine<br>53    | 131Xe<br>Xenon<br>54  |  |  |
| 133Cs<br>Caesium<br>55                                                     | 137Ba<br>Barium<br>56   | 139La<br>Lanthanum<br>57 | 179Hf<br>Hafnium<br>72  | 181Ta<br>Tantalum<br>73 | 184W<br>Tungsten<br>74   | 186Re<br>Rhenium<br>75   | 190Os<br>Osmium<br>76    | 192Ir<br>Iridium<br>77 | 195Pt<br>Platinum<br>78  | 197Au<br>Gold<br>79     | 201Hg<br>Mercury<br>80 | 204Tl<br>Thallium<br>81 | 207Pb<br>Lead<br>82     | 209Bi<br>Bismuth<br>83   | 210Po<br>Polonium<br>84  | 210At<br>Astatine<br>85 | 222Rn<br>Radon<br>86  |  |  |
| 223Fr<br>Francium<br>87                                                    | 226Ra<br>Radium<br>88   | 227Ac<br>Actinium<br>89  |                         |                         |                          |                          |                          |                        |                          |                         |                        |                         |                         |                          |                          |                         |                       |  |  |
| Key                                                                        |                         |                          |                         |                         |                          |                          |                          |                        |                          |                         |                        |                         |                         |                          |                          |                         |                       |  |  |

## Key

relative atomic mass

| $A_r$ | Symbol | Name | $Z$ |
|-------|--------|------|-----|
|-------|--------|------|-----|

atomic number